

MERN STACK WEB DEVELOPMENT COURSE

Build, Scale & deploy Modern Web Applications — From Beginner to Advanced

HTML

- What is HTML
- Structure of an HTML document
- HTML elements and tags
- Headings, paragraphs, line breaks
- Links and images
- Lists (ordered, unordered)
- Tables
- Forms and input types
- Semantic HTML (header, footer, section, article)
- Audio and video elements
- Iframes
- Accessibility basics (alt, label, aria)

CSS

- What is CSS
- Inline, internal, and external CSS
- Selectors and properties
- Colors, units, and measurements
- Box model (margin, border, padding, content)
- Text and font styling
- Backgrounds and gradients
- Display and visibility
- Positioning (static, relative, absolute, fixed, sticky)
- Flexbox
- Grid layout
- Responsive design and media queries

Bootstrap

- What is Bootstrap
- Why use Bootstrap
- Bootstrap installation methods
- Bootstrap project structure
- Bootstrap grid system
- Containers
- Rows and columns
- Responsive breakpoints

- Typography
- Colors and background utilities
- Spacing utilities (margin and padding)
- Display and visibility utilities
- Flex utilities
- Position utilities
- Buttons
- Button groups
- Alerts
- Badges
- Cards
- Images and image utilities
- Tables
- Forms
- Navbar
- Dropdowns

Module 3 — Web & Javascript Fundamentals

Web Fundamentals

- How the internet works (DNS, Browser, Servers)
- Client–Server architecture explained with diagrams
- HTTP vs HTTPS (methods, status codes, headers)
- Request–Response lifecycle
- Frontend vs Backend responsibilities

JavaScript Basics

- Variables (var, let, const)
- Data types (primitive & non-primitive)
- Operators (arithmetic, logical, comparison)
- Type coercion & truthy/falsy values

Control Structures

- Conditional statements (if, else, else if)
- Loops (for, while, do-while)
- switch case usage

Functions & Scope

- Function declarations vs expressions
- Arrow functions
- Parameters & return values
- Global vs local scope
- Hoisting behavior

Arrays & Objects

- Array methods (map, filter, reduce)
- Object creation & manipulation

- Nested objects

ES6+ Features

- let & const deep dive
- Destructuring (arrays & objects)
- Spread & rest operators
- Template literals
- Modules (import / export)

Asynchronous JavaScript

- Synchronous vs asynchronous execution
- Callbacks & callback hell
- Promises (then, catch, finally)
- async / await

Module 4 — Advanced JavaScript & Backend Readiness

Execution Context & Call Stack

- Global & function execution context
- Stack memory behavior
- How JS executes code internally

Closures

- Lexical scoping
- Practical closure examples
- Use cases (data privacy, counters)

Module 5 — Node.js Fundamentals

- Node.js overview & architecture
- NPM & package management
- Core modules: fs, path, http
- Creating basic servers
- Environment variables & configuration
- Understanding non-blocking I/O

Module 4 — Express.js & REST API Development

- Express.js setup & project structure
- Routing & controllers
- Middleware concepts

- REST API design principles
- Request & response lifecycle
- Error handling & validations
- Building scalable APIs

Module 5 — MongoDB & Database Design

- NoSQL vs SQL (conceptual clarity)
- MongoDB basics & CRUD operations
- Mongoose schemas & models
- Relationships (One-to-One, One-to-Many)
- Aggregation framework
- Indexing & performance optimization
- Data validation & security

Module 6 — React.js

- React overview & component architecture
- JSX, props & state
- Hooks (useState, useEffect, useContext)
- React Router & navigation
- Forms & validations
- API integration with backend
- Protected routes & authentication flow
- Performance optimization & best practices

Module 7 — React + Node Integration

- Connecting React frontend with Node APIs
- JWT authentication (login/signup)
- Role-based access control
- File upload & image handling
- Pagination, search & filters

Module 8 — SQL & Database Comparison

- SQL fundamentals
- Writing queries (CRUD)
- Joins & relationships
- SQL vs MongoDB — real-world use cases

Module 9 — TypeScript

- TypeScript fundamentals
- Types, interfaces & generics
- TypeScript with React
- TypeScript with Node & Express
- Writing scalable & safe code

Module 10 — AI Integration in Web Applications

- Introduction to AI for web apps
- Using AI APIs
- Prompt engineering basics
- Adding AI features to MERN projects

Module 11 — Project Planning & Implementation

- Planning application ideas and system architecture
- UI/UX design using **Figma** and database/API planning
- Developing main features using **MERN Stack**
- Final testing, debugging & publishing both projects on **GitHub**

Job Readiness & Industry Training (TOI)

- Domain review (JS, React, Node, Mongo)
- Resume & portfolio preparation
- GitHub & project presentation
- Mock interviews (technical & HR)
- Industry best practices & workflows

Skills to Master

- JavaScript (ES6+)
 - React.js
 - Node.js & Express.js
 - MongoDB & SQL
 - REST APIs
 - Authentication & Security
 - TypeScript
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- Deployment & Hosting

Tools to Master

- VS Code
 - Postman
 - Git, GitHub
 - MongoDB Atlas
 - Figma (UI Design)
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- AWS

Certification Outcome

Build **real-world full-stack projects**, deploy live applications, gain complete **industry-level MERN experience**, and become **job-ready as a Full-Stack Developer**.

Plan. Build. Deploy. — Become a Certified MERN Stack Developer